Open Source Intelligence (OSINT) Cyber Detective

Curtis Coleman
Assistant Professor
Cybersecurity Program Director
Office: PEC-224
Phone: 405.425.5472
Cell: 405-990-1842

OKLAHOMA CHRISTIAN UNIVERSITY

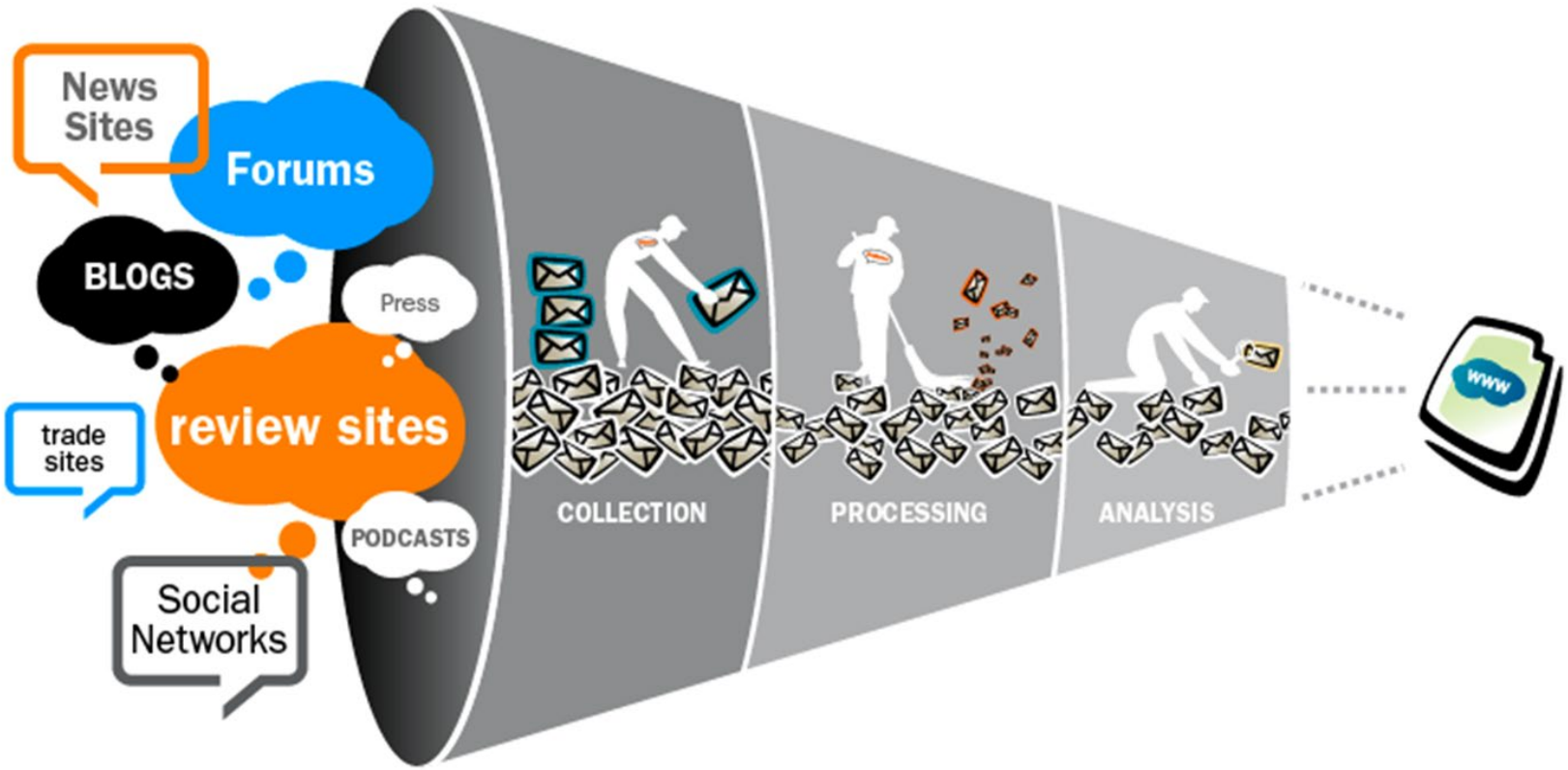
What is OSINT?

Open-Source Intelligence (OSINT) is intelligence collected from publicly available sources.

"**Open**" refers to overt, publicly available sources, as opposed to covert or clandestine sources.

WIKIPEDIA
The Free Encyclopedia

The OSINT Process



Source: https://blog.quantika14.com/blog/2016/07/15/que-es-osint/

General Searches

- Search to find addresses, phone numbers, or e-mail addresses
  - www.yahoo.com
  - www.infobel.com (International companies)
  - www.smartpages.com (businesses)
  - www.whowhere.com (address, phone)
  - http://search.lycos.com

Background Investigation

- Search to find additional information beyond address and phone numbers
  - www.theultimates.com/white
  - https://www.whitepages.com
  - https://www.beenverified.com
  - https://www.peekyou.com

Social Media sites to search:

Facebook, Instagram, LinkedIn, Twitter

Snapchat, Pinterest, Tumblr, Vine, Reddit

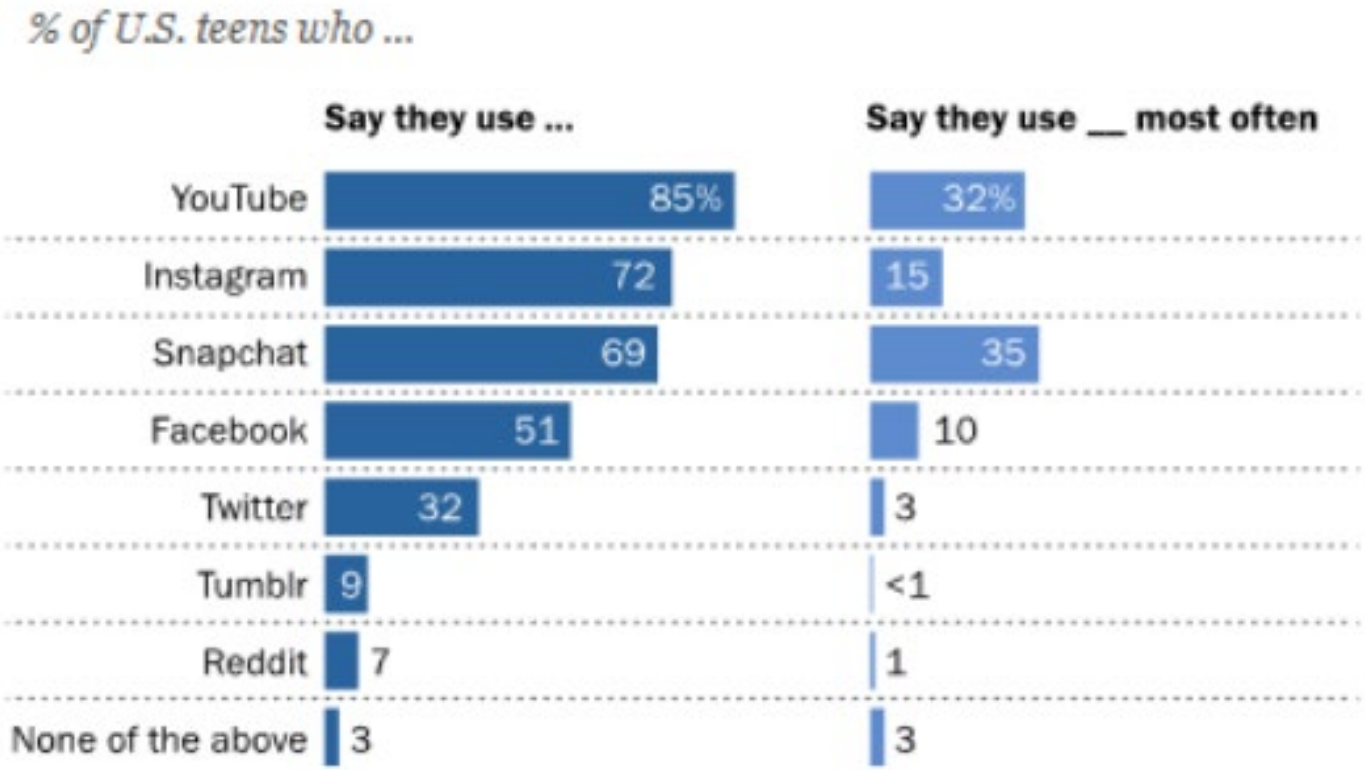
Kik, Pheed, Wanelo, GroupMe

Social Media Poll

Which of these Social Media do you use?

- YouTube _____
- Facebook _____
- Instagram _____
- LinkedIn _____
- Twitter _____
- Snapchat _____
- Pinterest _____
- Tumblr _____
- Reddit _____
- GroupMe _____

Social Media & Youths



Note: Figures in first column add to more than 100% because multiple responses were allowed. Question about most-used site was asked only of respondents who use multiple sites; results have been recalculated to include those who use only one site. Respondents who did not give an answer are not shown.
Source: Survey conducted March 7-April 10, 2018.
"Teens, Social Media & Technology 2018"

PEW RESEARCH CENTER

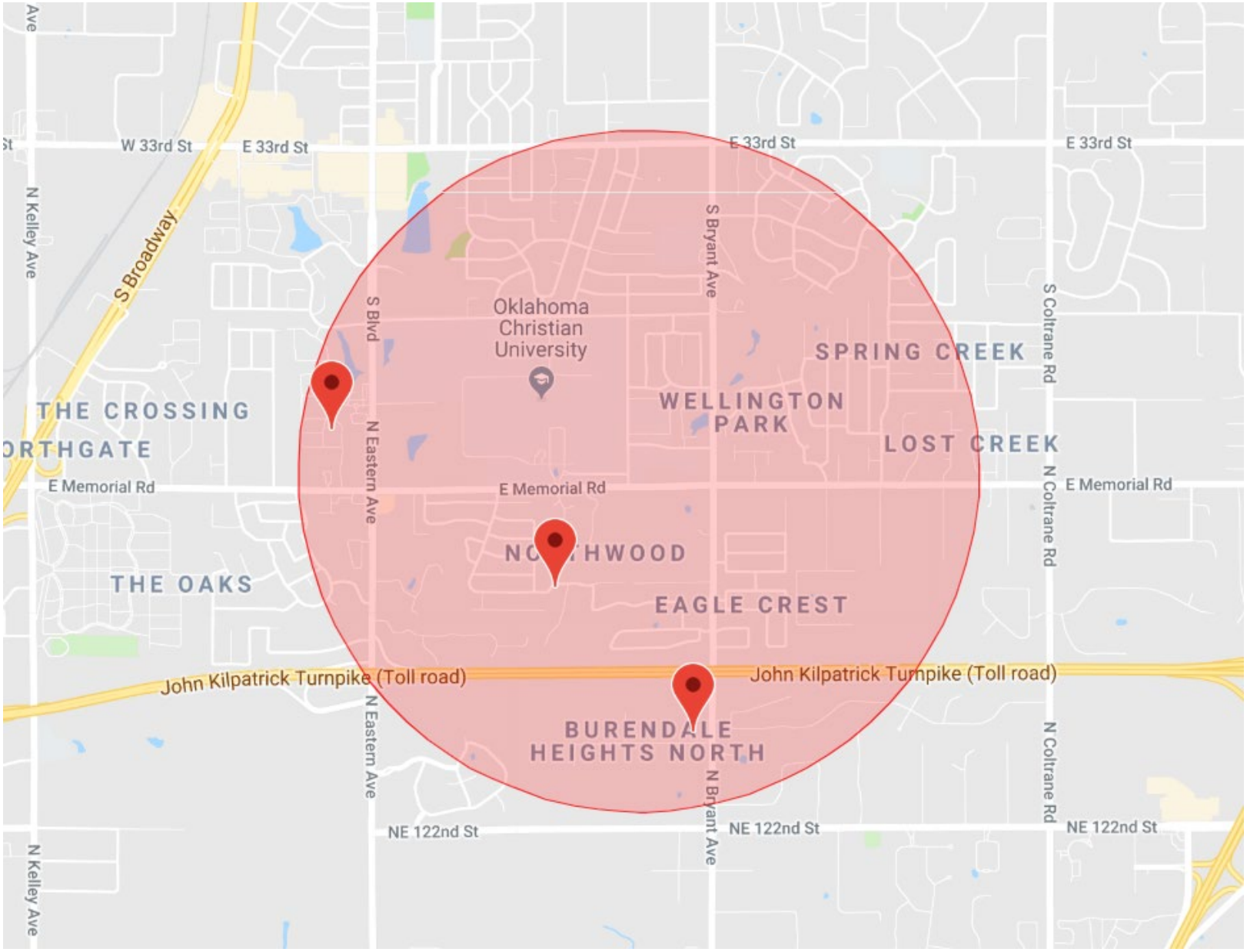
Court Records and Criminal Checks

- Sex offender registries
  - https://www.fbi.gov/scams-and-safety/sex-offender-registry
  - You may not be able to base employment decisions on certain information.
  - Check with an attorney first.

Oklahoma Sex Offender Registry

https://sors.doc.state.ok.us

- **Habitual** and **aggravated** offenders must register for this database for life.
- Other sex offenders are in the database for a period of **10 years** following the end of their sentence.



https://sors.doc.state.ok.us

Court Records and Criminal Checks

- Civil court records
  - Civil issues, as well as crimes, may make a person unsuitable for a particular job.
  - No centralized Web site for these issues.
  - Many states and Federal courts offer online records.

Oklahoma Court Records

Help | Contact Us | E-PAYMENTS | CAREERS

OKLAHOMA State Courts Network

HOME | COURTS | DECISIONS | PROGRAMS | NEWS | LEGAL RESEARCH | COURT RECORDS | QUICK LINKS

COURT SELECTION

County or Court:

All Oklahoma Courts*

SEARCH BY CASE NUMBER

Case Number:

Go

SEARCH BY PARTY

Last Name:

First Name:

Middle Name:

Born On or After Date:

Born Before Date:

- http://www.oscn.net/dockets/Search.aspx

Court Records and Criminal Checks

- Other resources
  - Pacer (Public Access to Court Electronic Records)
    - www.pacer.psc.uscourts.gov
  - Federal Bureau of Prisons records
    - www.bop.gov
  - Free Public Records
    - www.searchsystems.net

Usenet

- Newsgroups on many subjects.
- Use Google “Groups” option.
- Anyone can post anything .
- Search for potential employees.
- Can be an important investigative tool.
- Information must be verified elsewhere.

Use Yahoo to Search Usenet

Top OSINT Tools

Source: https://www.pinterest.com/pin/464434398934691 05/

1. Googledorks

- GHDB dorks can be used to reveal vulnerable servers on the Internet, to gather sensitive data, vulnerable files that are uploaded, sub-domains, and so on.
- Effective usage of GHDB can make the hacking process considerably easier.
- Exploit DB maintains a collection of googledorks under a section named GHDB.

Source: https://www.exploit-db.com/google-hacking-database

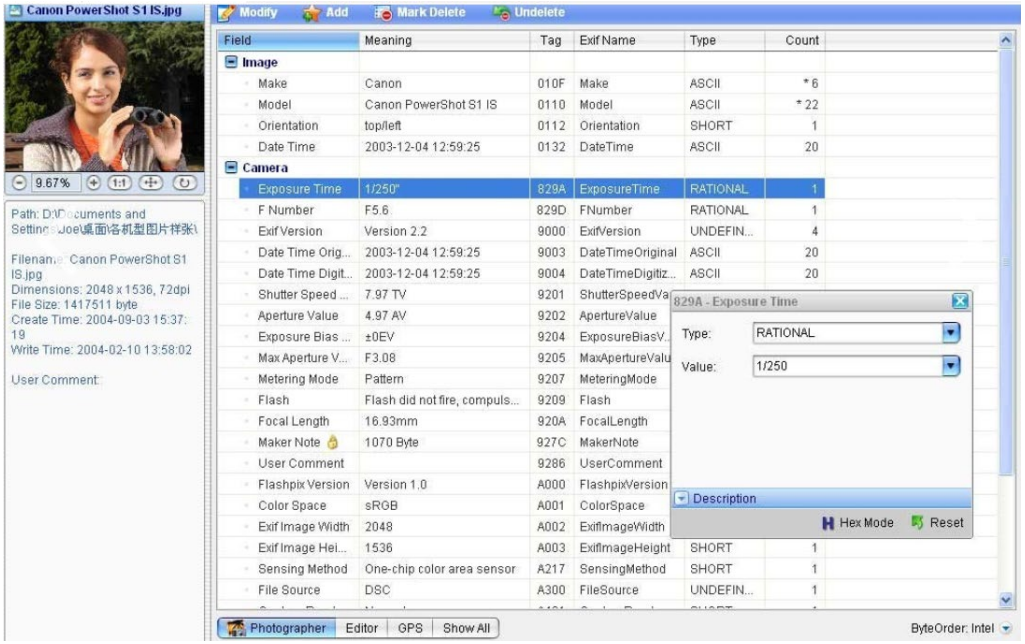
2. Maltego

- An extremely powerful OSINT framework, covering infrastructural reconnaissance and personal reconnaissance.
- Enables the gathering of sensitive data about the target organization, email addresses of employees, confidential files which are handled carelessly, internal phone numbers, DNS records, IP address information, geo location of the network, MX servers, and so on.

3. Shodan - Sentient Hyper Optimized Data Access Network

4. EXIF Data Viewers

- Smartphones and digital cameras use a standard to specify formats for images and sounds that are recorded using them.
- This standard is called the exchangeable image file format (EXIF).
- Geo-location in photo metadata can reveal the latitude and longitude

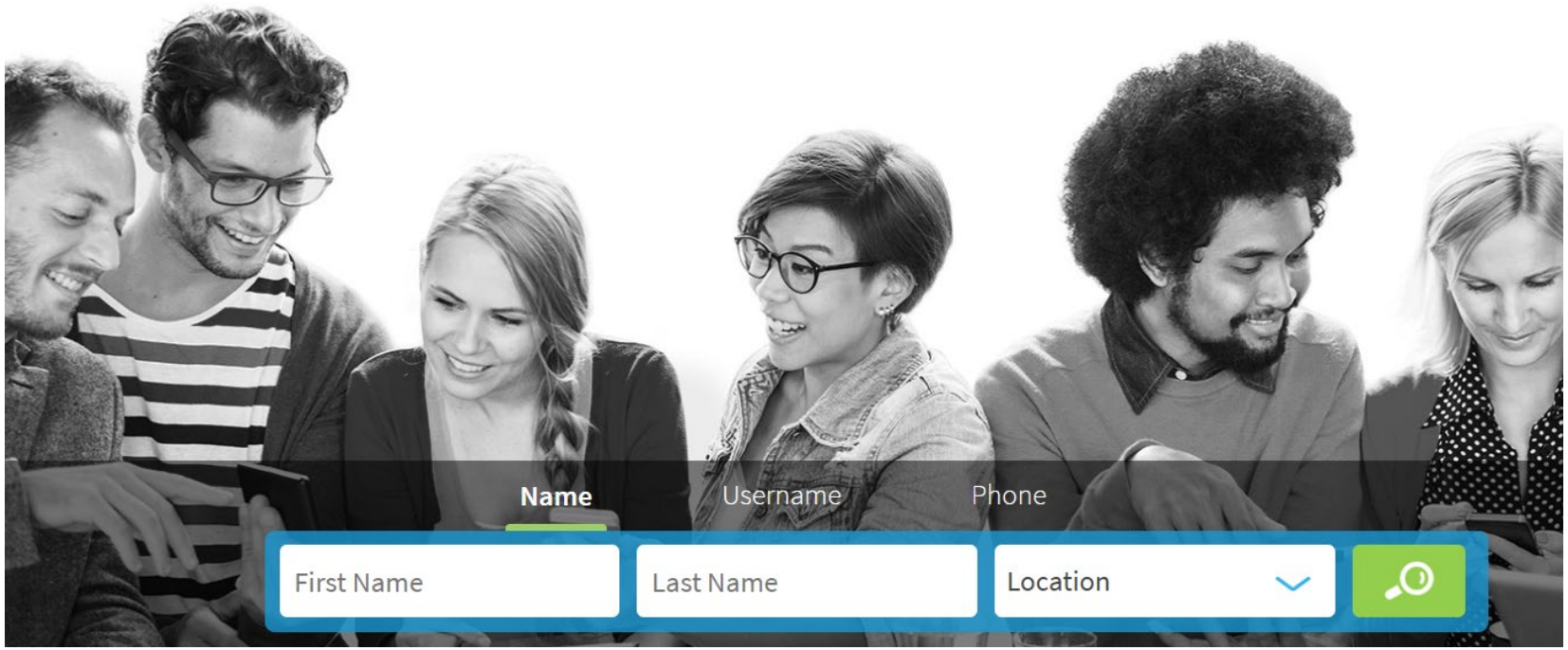


5. Cyberstalking tools for Reconnaissance

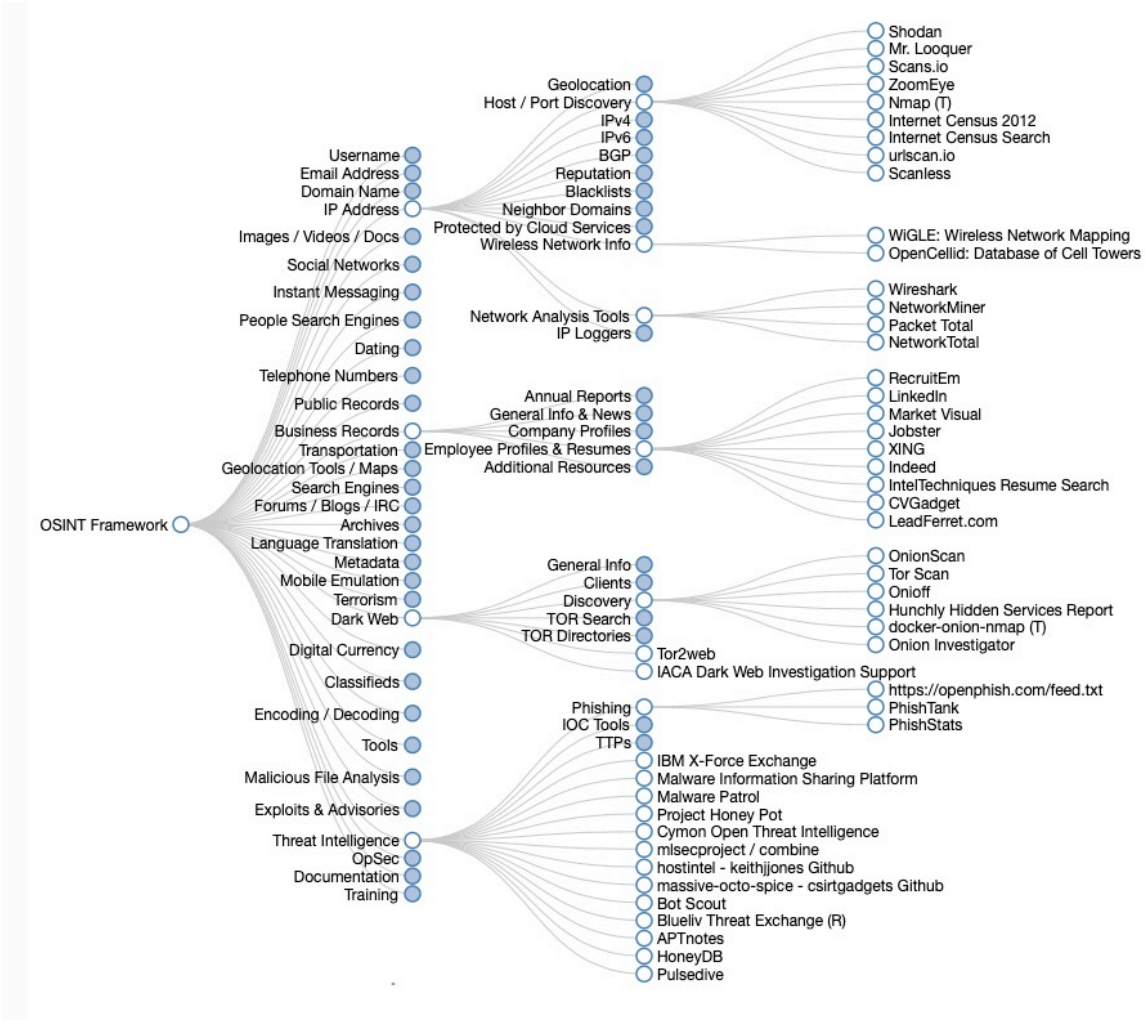
peekyou

Free People Search Made Easy

Find friends, relatives and colleagues across the Web.



6. OSINT Framework



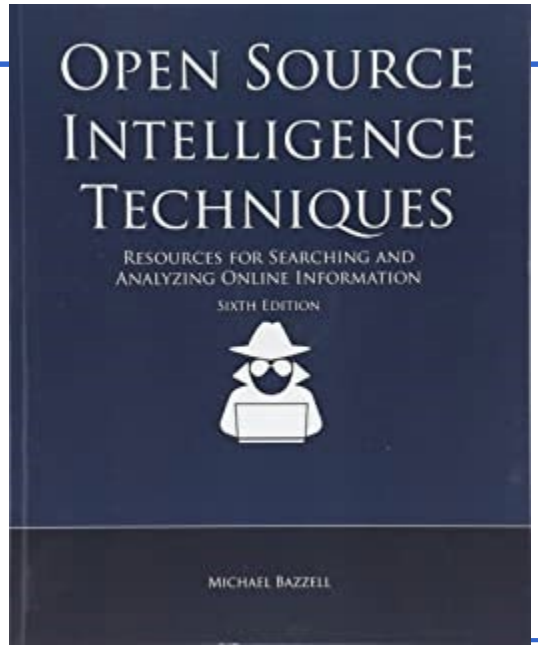
Source: https://osintframework.com/

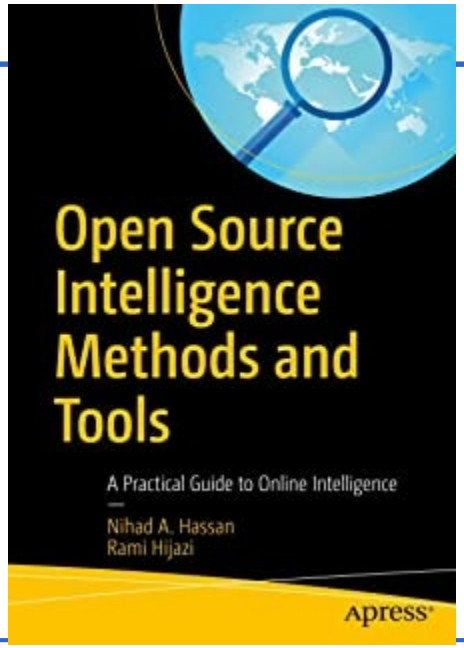
Can I get a job searching the Internet?

$97,500 / Year

The average **Source Intelligence Analyst** salary in USA is **$97,500** per year or **$50** per hour. Entry level positions start at **$35,750** per year while most experienced workers make up to **$137,341** per year.

Source:
http://neuvoo.com/salary/?job=Source+Intelligence+Analyst





1. Bazzell, Michael - *Open Source Intelligence Techniques: Resources for Searching and Analyzing Online Information*
2. Hassan and Hijazi - *Open Source Intelligence Methods and Tools: A Practical Guide to Online Intelligence*

QUESTIONS?

Curtis Coleman, CISSP, CISM
Assistant Professor
Cybersecurity Program Director
405-425-5472
Curtis.Coleman@oc.edu